

On-Device AI Models

Did you know your smartphone can run artificial intelligence models all on its own? With AI-optimized processors (like NPUs), your phone can handle tasks like image recognition or language translation without needing an internet connection. It's super handy for privacy and speed because everything happens locally.

Wondering how you can do the same? Follow these steps:

1. Get started by installing a local ML framework like TensorFlow Lite or PyTorch Mobile.
2. Download a pre-trained model for your phone. For example, grab an object detection model or a language processing model.
3. Use your phone's camera, microphone, or text input to gather real-time data.
4. Feed the data into the model, which processes it locally using your phone's AI chip.
5. Review the output directly on your phone – whether it's identifying an object or summarising text.



Precision Measurement Tools

Your phone is more than just a communication device – it's a pocket-sized measurement lab! With built-in sensors like the accelerometer, gyroscope, and magnetometer, you can measure tilt, vibration, or even magnetic fields. It's perfect for DIY experiments or engineering projects.

Want to try it out yourself? Here's how:

1. Access your phone's sensors via built-in diagnostic tools or developer-friendly APIs like Android Sensor APIs.
2. For tilt measurements, tilt your phone and observe the gyroscope data for angle changes.
3. To measure vibrations, place your phone on a moving surface and log accelerometer readings.
4. Record the data in a graphing app or spreadsheet for better visualisation.
5. Calibrate the sensor results for more accurate readings and apply them to your project.

DIY Spectroscopy

Ever thought you could turn your phone into a mini spectrometer? By using its camera and flashlight, you can analyse light spectra from different materials. It's a cool way to dive into optics and material science without fancy lab equipment.

Curious how it works? Let's break it down:

1. Grab a diffraction grating – an old DVD works great – and attach it in front of your phone's camera lens.
2. Shine a light (like your phone's flashlight) on the object you want to analyse.
3. Capture the spectrum that appears through the diffraction grating using your camera.
4. Observe the colours and patterns in the spectrum to identify material properties.
5. Document your findings and compare them with known spectral data for accuracy.

